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Challenge Report

EY challenge report

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1. Introduction

In Vietnam, rice crops are very important. It's crucial to classify and keep an eye on these fields. This project aims to do that by using satellite data and machine learning. We want to create a strong and reliable model that can accurately tell us which fields have rice crops. By using advanced technology and smart algorithms, especially machine learning, we hope to make farming better, protect the environment, help the economy, and contribute to scientific research. This project will show how these technologies can make a big difference in farming, food security, and sustainable practices in Vietnam.

2. Workflow Overview

This section provides an overview of our work's workflow and the underlying rationale. We will discuss the general steps involved, including **what** has been done, **how** it was executed, and provide **explanations & intuitions** for some important decisions. The details will be discussed in the next sections.

2.1. Workflow

The workflow of our work can be outlined as follows.

1. Data Visualization 3.1 : We began by visualizing the data to gain insights and understand the distribution of rice crops. This step may help to determine how to approach the data treatment.
2. Data Source Selection 3.2 : Given access to a platform with a vast amount of satellite data, we needed to select relevant data sources for our study. We chose Sentinel-1, Sentinel-2, Landsat, and Modis satellites because they cover the region of interest and provide suitable features for rice crop classification.
3. Data Preparation : After selecting the data sources, we proceeded to create the database. Each satellite provides different types of features over different time windows and resolutions. However, the information from satellites is not always clean, especially when faced with cloudy conditions. Thus, creating the database involved extracting satellite features while considering these factors 3.2.
4. Model Architecture 4.1 : After experimenting with various machine learning approaches, we arrived at the following structure (schema resuming this part3) :
 - Encoding : To handle the information gathered from each satellite over a time window, we employed a technique called Gated Recurrent Unit (GRU) [1]. This process resulted in four compressed vectors that capture the temporal evolution of the features from the 4 satellites.
 - Separation : We utilized a discriminator, specifically an Artificial Neural Network (ANN) [2], to classify the compressed data obtained from the encoding step. The discriminator aids in making classification decisions.
5. Model evaluation 4.2 : Now that we have the model architecture we will use a method called "k-fold cross-validation with model ensembling." It is a way to evaluate how well a machine learning model performs on a given dataset. Here's how it works (schema resuming this part4) :
 - The dataset is divided into 2 subsets (**train-val** data & **test** data), the **train-val** subset is divided into smaller groups called "folds." In our case 9 folds.

2. Workflow Overview

- The model is trained and tested 9 times. Each time, a different fold is used as the "validation" data, and the remaining 8 folds are used for training.
- After each training, the model's performance on the test data is evaluated. The best performing model on the validation data is saved each time.
- In the end, we have 9 models, each trained on a slightly different subset of the data.
- The final prediction is made by combining the predictions from the 9 models. We use a voting system, where the prediction that most of the models agree on becomes the final prediction.
- we then test the model on the subset **test**

2.2. Intuitions

Why these satellites ?

- Sentinel 2 & Landsat : These satellites give us optical images, which means they capture images using visible light. While these images can be affected by cloud cover, we found a clever way to deal with this challenge. We realized that if one satellite captures a cloudy image for a specific location, the other satellite might have a clear image of the same location at that time. By combining the data from both satellites, we can effectively tackle the problem of cloud interference and obtain a more complete picture of the rice crops' distribution.
- Sentinel 1 : This satellite provides high-resolution radar images. Unlike optical satellites, Sentinel 1's radar technology allows it to capture images even in cloudy or unfavorable weather conditions. This is crucial because it ensures a continuous and reliable source of data throughout the time window we're interested in. So, we can gather valuable information about the rice crops regardless of the weather.
- Modis : Modis, on the other hand, provides us with a broader perspective. Although its resolution is not as high as the other satellites, it covers a wider area. This means we can gather contextual information about the region surrounding our points of interest. For example, if our point is in a coastal area or in the middle of a dense forest, Modis helps us understand the environmental context. This context plays a crucial role in accurately classifying the rice crops.

By utilizing these satellites together, we can leverage the strengths of each one. Sentinel 1 ensures consistent data collection, Sentinel 2 and Landsat complement each other to overcome cloud cover issues, and Modis adds valuable context to our analysis. This combination of satellite data enables us to gain insights into the distribution and characteristics of rice crops more effectively.

Why encode the data from each satellite separately and not combine them all into one vector ?

One challenge we encountered with satellite data is missing data. Different satellites provide varying amounts of clean data at specific points in our time window, creating an offset in the data evolution between satellites. This can have significant consequences during the learning phase.

To handle this challenge, we chose to encode the data from each satellite separately. By doing this, we ensure that the data is properly represented based on its specific interval of existence. This approach avoids introducing irrelevant information and maintains the integrity of each satellite's data.

By encoding the data separately for each satellite, we capture their unique characteristics and evolution accurately. This allows our model to learn and interpret the data correctly, considering the different time

intervals and data availability from each satellite.

Why this method in model evaluation ?

K-fold cross-validation with model ensembling is particularly useful when working with **small datasets**. Small datasets often pose a higher risk of overfitting, where the model fits the noise in the data instead of the underlying patterns. By using k-fold cross-validation, the model is trained and evaluated on different subsets of the data, reducing the chance of overfitting and providing a more accurate performance estimate. By combining the predictions of multiple models trained on different subsets of the data, the ensembled model can potentially achieve better performance. This approach leverages diverse perspectives learned from different subsets, enhancing the model's ability to generalize and make **robust** predictions

3. Data exploitation & preparation

3.1. Data exploitation

3.1.1. Interactive map :

An interactive map was developed using Leaflet as part of the project to visualize the given coordinates and understand their spatial distribution. The map allows for spatial exploration, pattern identification, and visualization of the geographic locations associated with rice fields in Vietnam. However, it is important to note that the geographic location itself was not directly used for model learning. Instead, features extracted from satellite data, such as spectral information and vegetation indices, were used as inputs for the machine learning model. The interactive map serves as a complementary tool to enhance spatial awareness and aid in data mining, ensuring the model is trained based on relevant features rather than geographic position alone. Additionally, the map helped in selecting an appropriate degree box size for data extraction from satellite imagery by observing the coordinate distribution. The interactive map allows users to zoom, scroll, and explore the data in detail, facilitating the identification of clusters based on geographical proximity.

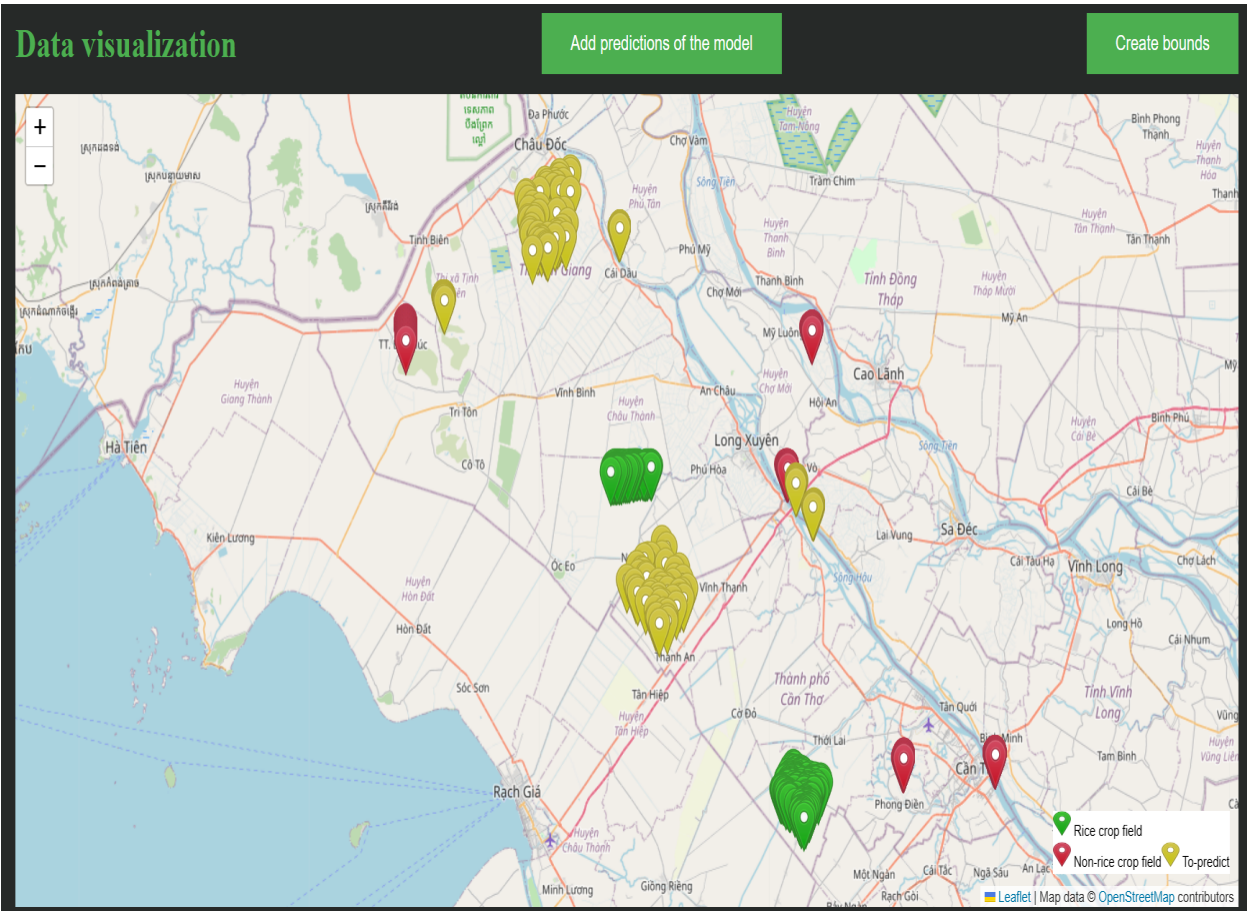


FIGURE 1 – index page of the web interface

3.1.2. Size box :

By clicking on the "Add bounds" button on the web interface, users can navigate to the second page, which offers additional functionality for exploring specific markers on the map. Upon zooming in on a particular marker of interest, users can click on it, triggering the appearance of a corresponding bounding box. This bounding box represents the specific region or area that was considered when extracting satellite data for analysis. The inclusion of these boxes helps users understand the spatial context of the extracted data and provides a visual reference for the model's training and prediction process. This feature allows users to delve deeper into the individual markers, view the corresponding bounding boxes, and gain insights into the specific areas that influenced the model's classification results. The box size was chosen based on the resolution of satellites, we wanted it to be large enough to capture the complete extent of the feature (about 5*5 pixels), but relatively small comparing to the surface of the rice crop field.

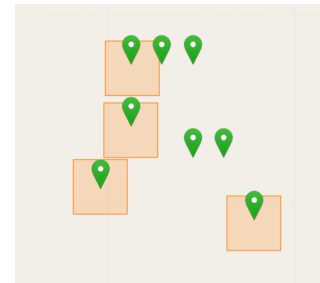


FIGURE 2 – Addbox page of the web interface

3.2. Data extraction

During the data extraction phase, features were extracted from four satellite products :

- Sentinel 1 Radiometrically Terrain Corrected (RTC)[3] :
- Sentinel-2 and Landsat [4][5]
- MODIS Vegetation Indices 16-Day :[6]

By extracting features from multiple satellites, each contributing unique insights, the project obtained a comprehensive dataset that captured various aspects of the rice crop fields in Vietnam. Furthermore, our approach went beyond merely extracting data within the chosen degree size box. We aimed to incorporate additional local measures corresponding to the specific coordinates of interest. By including these local measures, we obtained a more comprehensive representation of the field, taking into account localized characteristics that could contribute to the classification task. Additionally, the time window that was considered is from 01-09-2021 to 20-06-2022. Here is a table that summarizes the extraction phase :

Satellite	Resolution	Extracted bands	degree size box used	Aggregation method(s)
Sentinel 1	20m pixel	VH : vertical transmit, horizontal receive, VH : vertical transmit, horizontal receive	0.0008	mean, median
Sentinel 2	10m pixel	B02 : Blue, B03 : green, B04 : red, B08 : nir	0.0008	median
Landsat	30m pixel	TMB1 : blue, TMB2 : green, TMB3 : red, TMB4 : nir08	0.0008	median
MODIS	500m pixel	Surface Reflectance Bands : MIR, NIR, red, blue, EVI, NDVI	the corresponding pixel	none

3.2.1. Sentinel 1 extracted data

In our data extraction process, we made a conscious effort to leverage the abundant data available from Sentinel-1. One of the key advantages of using Sentinel-1 data is its resilience to cloud cover. Our aim was to gather comprehensive information about specific features using measurements such as VV and VH. To achieve this, we've adopted a multi-faceted approach. Firstly, we've focused on local information by retrieving the values corresponding to specific coordinates, enabling us to obtain detailed insights about individual pixels (features vv-min-res and vh-min-res). Additionally, we utilize a degree-sized box to capture a more global perspective. This allowed us to gather a broader understanding of the surrounding area. To process the data within the box, we've employed different aggregation methods such as calculating the mean and median. The mean provides an average value (features vv and vh), while the median offers a central tendency measure less influenced by outliers (features vv-median and vh-median). By combining local and global information and utilizing various aggregation techniques, we've gained a comprehensive and nuanced view of the features observed in the Sentinel-1 data extraction process.

3.3. Data preprocessing

Preprocessing the data to reduce the number of features is crucial for efficient model training, improved performance, and practical deployment. Having a large feature space can lead to computational complexity and longer training times. By selecting a subset of relevant features, we focus the model on the most informative aspects of the data, enhancing its predictive power. In addition to the extracted data, we further enhanced the feature set by incorporating vegetation indices that have proven to be effective in this context, such as the Normalized Difference Vegetation Index (NDVI), Enhanced Vegetation Index (EVI), and Soil-Adjusted Vegetation Index (SAVI). These indices capture specific characteristics related to vegetation health and vigor, providing valuable insights for classifying rice crop fields. Feature selection, including these indices, aids interpretability and computational efficiency, making the model more practical for real-world applications. Techniques like correlation analysis help identify redundant features and ensure a compact feature set that captures essential information for accurate predictions.

3.3.1. Vegetation indices :

These new indices were defined based on the data that were already extracted from satellites :

Index	Formula	Relevance to rice crop fields
The Radar Vegetation Index (RVI)	$RVI = 4 * \sqrt{1 - \frac{VV}{VV+VH}} * \frac{VH}{VV+VH}$	vegetation structure and density
The Normalized Difference Vegetation Index (NDVI)	$NDVI = \frac{NIR-Red}{NIR+Red}$	vegetation density, vegetation cover, and photosynthetic activity
Soil-Adjusted Vegetation Index (SAVI)	$SAVI = 2 * \frac{NIR-Red}{NIR+Red+1}$	soil reflectance
The Enhanced Vegetation Index (EVI)	$EVI = 2.5 * \frac{NIR-Red}{NIR+6*Red-7.5*Blue+1}$	vegetation density and reduction of atmospheric influences

3.3.2. Correlation between features :

Given that the time series length for each satellite varied, we adopted a strategy to calculate the correlation between features independently for each satellite's time series. This approach was chosen to address the potential variations in data patterns and characteristics captured by different satellites over time. By calculating correlations within each satellite's time series, we can effectively identify and retain features that exhibit consistent relationships and patterns within the specific temporal context of each satellite.

Satellite	Correlated features	correlation value	kept feature(s)
Sentinel 1	RVI for different aggregation methods	0.99	RVI_median
	vv_median, vh median and vv_mean,vh_mean	0.97	vv_mean, vh_mean
Sentinel 2	Green, blue and red bands	0.82	green band
Landsat	Green, blue and red bands	0.93	green band
MODIS	blue and red bands	0.98	blue band
	NDVI and EVI bands	0.99	NDVI band
	MIR and NIR bands	0.94	NIR band

4. Model description

The language used in the whole programming part is Python, and the framework used in the learning part and for building models is PyTorch[7]

4.1. Model architecture

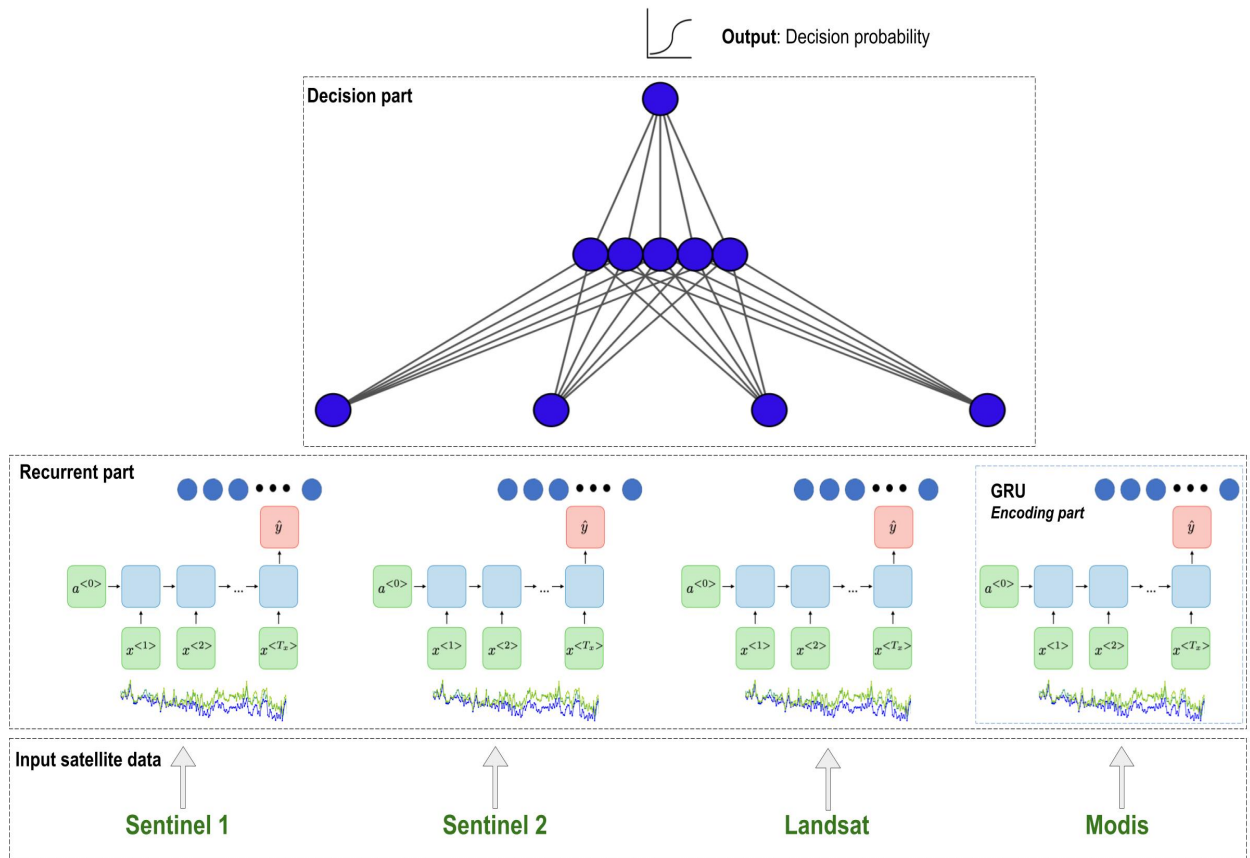


FIGURE 3 – Model architecture

The architecture3 of the model can be divided into two main parts :

Recurrent part

In the recurrent part, four Gated Recurrent Unit (GRU) layers are used to encode data from each satellite. The hyperparameters used for the GRU layers are : number of layers = 1, hidden size = 64, and dropout probability = 0.4. Each GRU layer processes the data from one satellite, capturing temporal patterns and encoding the information into vectors.

Decision part

In the decision part, the outputs from the recurrent part, which are four vectors of size 64, are concatenated. This concatenated vector serves as input to a two-layer Neural Network. The first layer has a size of 32, and the second layer has a size of 1. The output of this network is a real number representing the likelihood of a point belonging to a rice crop. To obtain the final probability of a point being part of a rice crop a sigmoid function is applied to the output.

During training, a learning rate of 0.001 is used, and the model is trained for 200 epochs.

4.2. Model evaluation

The method used to evaluate the performance of our model is "k-fold cross-validation with model ensembling". The first step is to divide the dataset into training and test sets, the training set is then also divided into 9 equal folds. Then, the model is trained 9 times, each time using a different fold as the validation set and the remaining 8 folds as the training set. In each iteration, the best performing model on the validation set is saved, resulting in 9 models, each trained on a slightly different subset of the data. In the final step, these 9 models are used to make predictions on the test set. The predictions from each model are then combined using a voting mechanism to produce the final prediction for each data point in the test set.

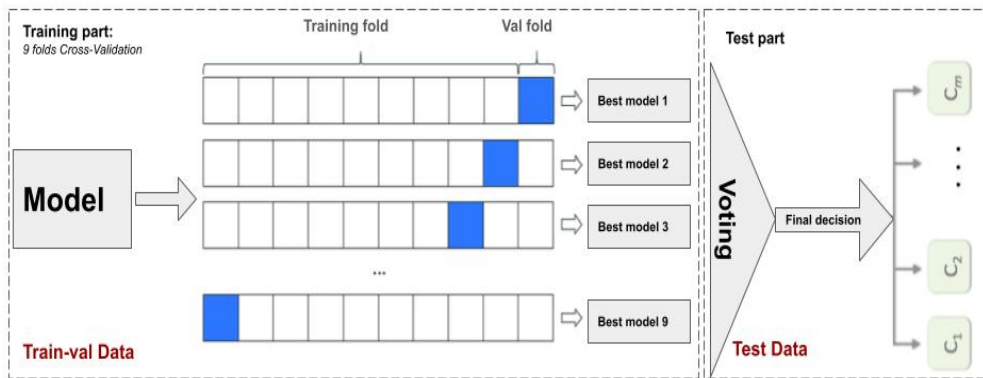


FIGURE 4 – Model evaluation

5. Result & analysis

After training our model, we've obtained an accuracy of 1.00 on the test set (a set that he hasn't seen before during the training process), this set was composed of 60 examples and was extracted from the training set before the beginning of the training process.

5.1. Technical Analysis

After submitting the model's results, an impressive score of 0.99 was achieved. To explore the spatial distribution of the model's predictions, a new button, "Add predictions of the model," was added to the index page. Clicking on this button allows users to view the model's predictions. Interestingly, the evaluation data exhibited distinct clusters, with each cluster corresponding to a specific prediction class (either all elements in the cluster were classified as rice crop fields or non-rice crop fields). However, it was observed that one rice crop field belonged to a cluster of non-rice crop fields, indicating a possible misclassification.

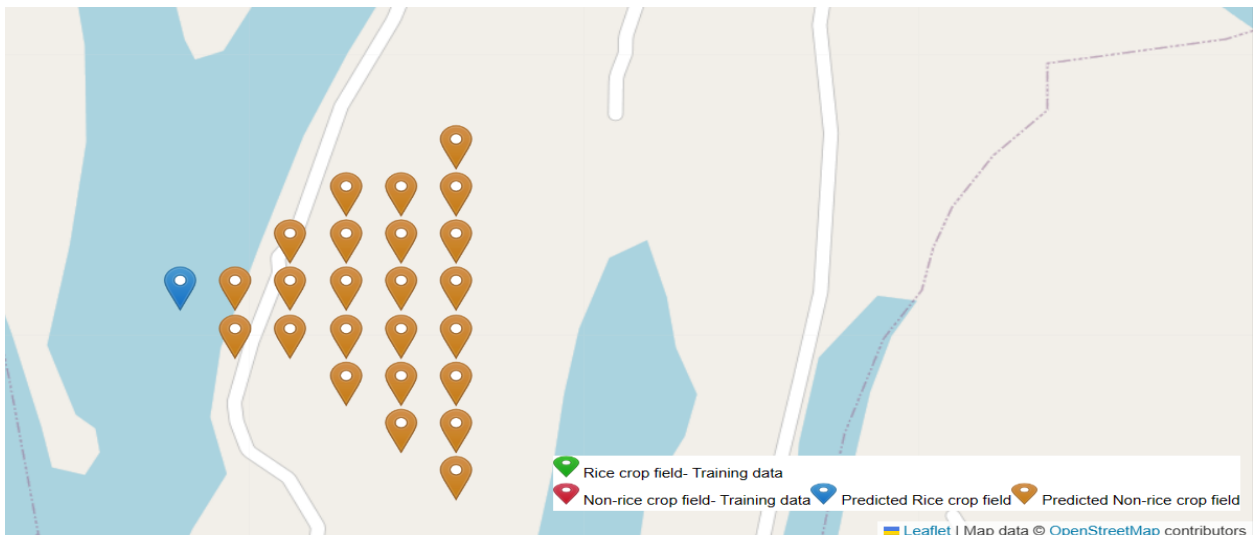


FIGURE 5 – Possible misclassified field

To investigate this anomaly, the field's features were analyzed. Further examination using Google Maps revealed that the field, despite being located near a river, appeared to be vegetation land. This discrepancy prompted a deeper analysis to understand the factors contributing to this misclassification. To narrow down the analysis of features, we focused specifically on the VV and VH bands, given their potential significance in explaining the misclassification. But first, let's see the overall evolution of bands VV and VH for rice crop fields.

5.1.1. Analysis of the evolution of polarization states VV and VH for rice crop fields :

In the context of radar remote sensing, the term "backscatter" refers to the portion of the radar signal that is reflected back to the sensor after interacting with the Earth's surface or objects on the surface. Backscatter is an essential measurement in radar remote sensing as it provides information about the properties of the

target or scene being observed. In the case of rice plants, the backscatter evolution over time is of particular interest. As rice plants grow, their interaction with the radar waves changes, resulting in variations in the backscatter signal. Initially, when the rice seedlings are submerged in water (November-December), the backscatter is minimal due to the specular reflection on the water surface. As the plants emerge and grow above the water surface, the backscatter increases due to the mechanisms of double bounce and multiple bounces. During the later growth stages, the backscatter decreases as the mechanism of volume scattering dominates.

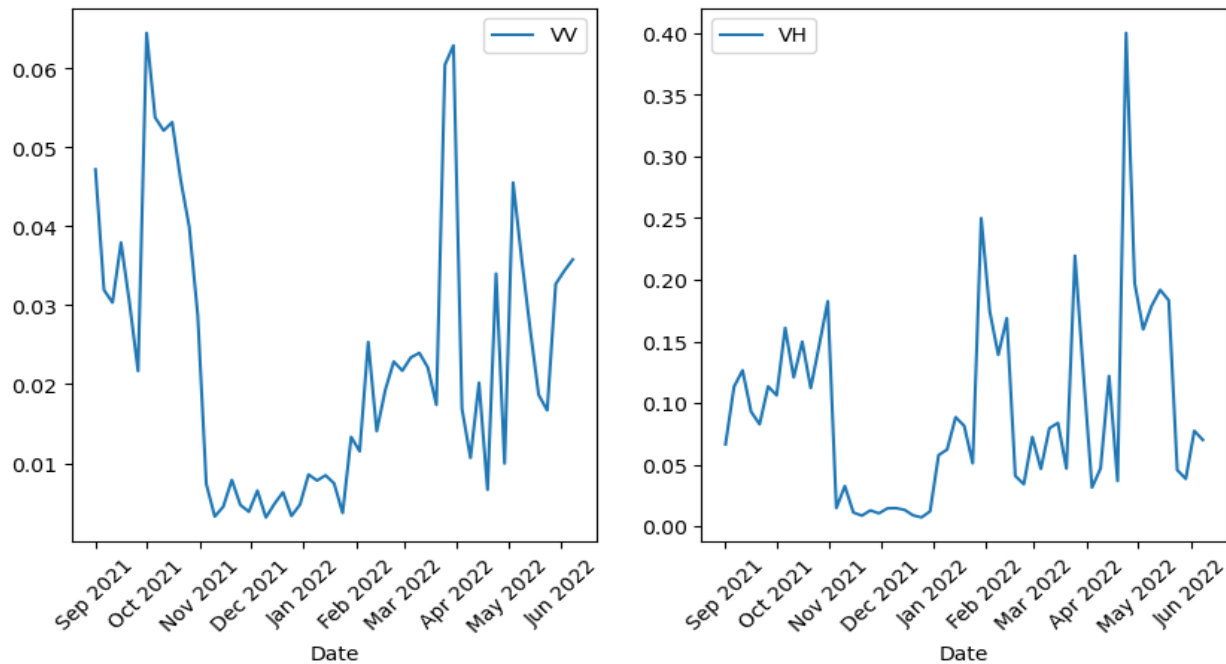


FIGURE 6 – Evolution of backscatter for rice crop fields

5.1.2. Plots of the misclassified field

We can see that the plot of, what we believe is, the misclassified field have a similar evolution 7 . Thus we think that this field might correspond to other plants that may have similar characteristics. In fact, other vegetation types or land cover features with, such as certain wetland areas or flooded vegetation, may exhibit comparable backscatter patterns. Therefore it can be sometimes very challenging to classify perfectly vegetation lands.

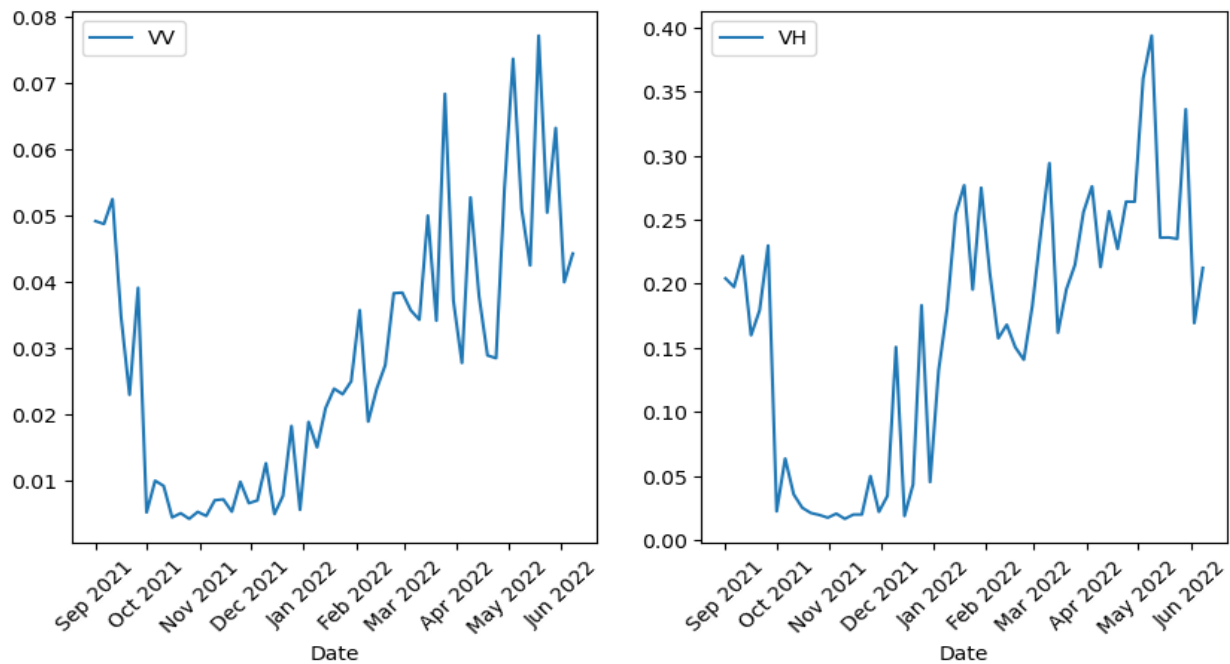


FIGURE 7 – Evolution of backscatter of the misclassified field

5.2. Environmental Analysis

Our model for identifying rice crop fields has positive effects on the environment. It helps manage agricultural land better by accurately finding rice fields. Among these effects :

- Using satellite data in the model allows us to monitor rice fields without disturbing the environment. We don't need to do extensive surveys or physical inspections, which can disrupt nature. It also helps monitor remote or inaccessible areas, giving us a better understanding of rice farming and its impact on the surrounding ecosystems.
- The model's ability to distinguish between rice fields and other types of land cover helps identify changes in land use and possible encroachments on natural habitats. This information is valuable for managing and conserving land, enabling us to intervene promptly and protect ecologically sensitive areas.

6. Encountered problems

In some phases of our solution, we encountered various challenges and potential problems. Here are some issues that we have faced :

- Data source detection : Selecting suitable data sources involves considering factors such as coverage, availability, and the relevance of features for rice crop classification. In addition of the 3 satellites that were proposed in the comprehensive guide of the challenge, we've decided to add data from satellite MODIS.
- Model architecture : In the beginning, we used LSTM (Long Short-Term Memory) as the unit in the recurrent part of our model because we believed its complex architecture could help encode complex information. However, we discovered that GRU (Gated Recurrent Unit) models were more efficient for our specific case. LSTM tends to be more efficient when dealing with very long sequences, which is not our scenario. Furthermore, we experimented with adding attention mechanisms to our models. However, we found that these mechanisms didn't significantly contribute to the overall performance and value of our classification model. Additionally, tuning the parameters of our model posed a considerable challenge.
- Model evaluation : Sometimes our model gives very different results when retraining it, we wanted our model to make robust predictions, the solution was to implement an ensemble method, specifically the "voting" technique.

7. Conclusion

In conclusion, our rice crop fields classification model has demonstrated strong potential in accurately categorizing rice cultivation areas. Through a combination of data visualization, careful data source selection, and effective data preparation techniques, we achieved reliable results. The integration of GRU and ANN models, along with ensemble methods like voting, improved the robustness and stability of our predictions. By leveraging satellite data from various sources and considering factors like cloud cover, we maximized the model's performance. While our approach has shown promise, there are still opportunities for further refinement and exploration. Overall, our model contributes to the larger goal of achieving food security and offers valuable insights for promoting sustainable agriculture, and supporting environmental conservation efforts.

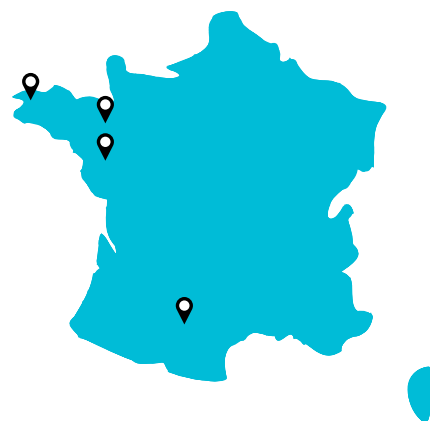
Références

- [1] [Gated recurrent unit](#)
- [2] [Artificial neural network](#)
- [3] [Sentinel 1 Radiometrically Terrain Corrected \(RTC\)](#)
- [4] [Sentinel-2 Level-2A](#)
- [5] [Landsat Collection 2 Level-2](#)
- [6] [MODIS Vegetation Indices 16-Day \(500m\)](#)
- [7] [Pytorch](#)

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